INF2009: Edge Computing & Analytics
Project Proposal Report

This report requires you to respond to a total of six questions, which are categorized under Section I of the final report. Ensure that you change the title of report to “Team_No_xx_Proposal” while uploading on Dropbox.

The purpose of this report is to submit the **Introduction, Problem Statement and Objectives** for your project. The report must answer the below questions:

1. What specific issue or problem has been identified that requires a solution?
2. Why is your topic important or relevant to industry, academia, or society?
3. What is the potential impact of addressing this challenge?
4. What specific gap or challenge in the field does this work address?
5. What are the main goals of this project or research?
6. What specific outcomes are expected by the end of this study?
7. Do include images/diagrams to illustrate your ideas.

1. **What specific issue or problem has been identified that requires a solution?**

   GymPulse: A Multi-Modal Edge Analytics System for Smart Condo Facilities

   In a private condominium gym, residents are not aware of the **actual crowd level** or **equipment availability** until they physically arrive. This leads to wasted trips, waiting time to use machines, and a poorer workout experience during peak hours. Condominium staff and property representatives also do not have **real-time utilization visibility**, because manual counting and log-based methods do not accurately reflect the true number of residents inside the gym at any given moment.
   Therefore, there is a need for a **real-time monitoring solution** that estimates gym occupancy and equipment usage using **a webcam + 4 PIR sensors** (2 at the entry for in/out counting, 2 at equipment zones for usage detection), and displays the results on a **web dashboard** with notifications.

2. **Why is your topic important or relevant to industry, academia, or society?**
   This project is relevant because it reflects the growing adoption of **real-time sensing and edge analytics** in smart buildings to improve user experience and optimize facility operations. For residents, live gym crowd status and equipment usage information helps them choose the best time to visit and reduces congestion. For condo management, real-time usage insights support better planning for cleaning, maintenance, and facility improvements. Academically, this demonstrates how edge computing can integrate **multiple distributed sensor nodes** (camera + PIR) with an **edge server** to deliver low-latency, privacy-aware analytics and a live dashboard for decision-making.

3. **What is the potential impact of addressing this challenge?**
   If this issue is addressed, residents can monitor live **gym crowd status** (e.g., Low/Medium/High) and see whether selected equipment zones are **in use or available**, allowing them to plan visits more efficiently and avoid peak congestion. For management, the system provides **historical trends** for both occupancy and equipment utilization, enabling better scheduling for cleaning and maintenance and supporting longer-term capacity planning. Additionally, the system can send **notifications** (e.g., crowd level drops, equipment becomes available, peak-hour alerts), improving convenience for residents and operational responsiveness for staff.

4. **What specific gap or challenge in the field does this work address?**
   Many basic gym occupancy solutions rely on a single data source such as camera-only analytics or access logs, which can be unreliable due to occlusions, lighting changes, tailgating, or inaccurate logging. This project addresses that gap by combining two complementary sources:

   - **Webcam analytics** for visual people estimation / crowd density classification
   - **Entry PIR sensors (2 units)** to detect movement patterns and estimate **in/out flow**, providing continuity even when camera accuracy fluctuates
   - **Equipment PIR sensors (2 units)** to detect **equipment zone usage**, adding a second dimension beyond “how crowded” → also “what is being used”

   By fusing these signals at the edge, the system improves reliability and provides more meaningful facility utilization insights than occupancy alone.

5. **What are the main goals of this project or research?**
   The primary goal is to develop a **Condo Gym Crowd + Equipment Monitoring Edge Analytics System** with:

   Sensor inputs

   1. **Webcam** for people/crowd density estimation
   2. **2× PIR sensors at the main entry** to estimate resident flow (entry/exit events)
   3. **2× PIR sensors at gym equipment zones** to detect equipment usage activity

   Deployment architecture (3× Raspberry Pi)

   - **RPI #1 (Sensor Node A):** Webcam + 2× entry PIR sensors (captures visual crowd estimate + entry flow events)
   - **RPI #2 (Sensor Node B):** 2× equipment PIR sensors (captures equipment usage activity)

- **RPI #3 (Edge Server):** Collects data from RPI #1 and #2, performs basic fusion/smoothing, updates the **Web Dashboard**, and triggers **notifications**

The system aims to produce a reliable, real-time “Gym Status” and equipment usage indicators with low latency and privacy-conscious edge processing.

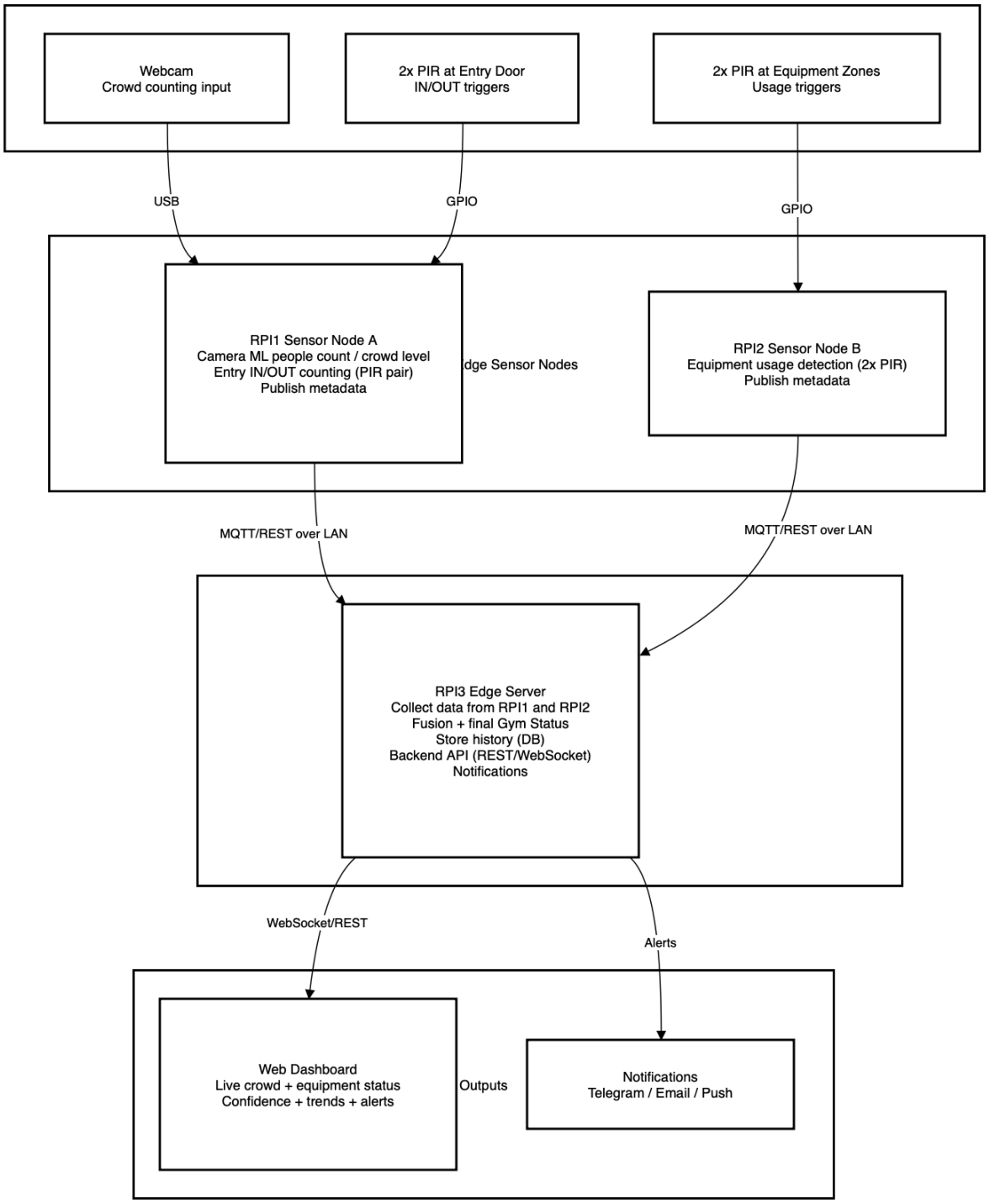
6. **What specific outcomes are expected by the end of this study?**
   By the end of this project, the expected outcome is a working prototype that integrates **webcam + 4 PIR sensors**distributed across **two sensor RPIs**, with a **third RPI acting as the edge server**. The system should estimate gym occupancy and crowd level in real time, detect equipment usage activity, and display both **live status** and **historical trends** on a dashboard.

   A successful demo will show multiple people entering/leaving the gym and interacting with equipment zones, where:

   - entry PIR signals reflect changing flow,
   - webcam analytics supports crowd estimation,
   - equipment PIR sensors indicate active usage,
   - the edge server updates the dashboard quickly and sends notifications based on defined triggers (e.g., crowd level changes, equipment becomes idle).

7. **Do include images/diagrams to illustrate your ideas.**

Sample diagram:
https://drive.google.com/file/d/1evuzVe0O_6o80wU6Hufbe9pxCe-Ehdnz/view?usp=share_link


Webcam
Crowd counting input
2x PIR at Entry Door
IN/OUT triggers
2x PIR at Equipment Zones
Usage triggers
USB
GPIO
GPIO
RPI1 Sensor Node A
Camera ML people count / crowd level
Entry IN/OUT counting (PIR pair)
Publish metadata
dge Sensor Nodes
RPI2 Sensor Node B
Equipment usage detection (2x PIR)
Publish metadata
MQTT/REST over LAN
MQTT/REST over LAN
RPI3 Edge Server
Collect data from RPI1 and RPI2
Fusion + final Gym Status
Store history (DB)
Backend API (REST/WebSocket)
Notifications
WebSocket/REST
Alerts
Web Dashboard
Live crowd + equipment status
Confidence + trends + alerts
Outputs
Notifications
Telegram / Email / Push

Gym Images:

Legend

① Webcam (people counting / crowd level)
② mmWave sensor (presence/motion verification at entrance)
③ Edge device (on-site inference + dashboard server)
④ Temperature sensor (Phase 2 enhancement)

Privacy note: "System outputs only occupancy count + status; no identity recognition."

Figure 1. *Deployment concept at a condominium gym entrance. The webcam (①) provides real-time crowd estimation within the counting ROI, while the mmWave sensor (②) verifies presence and entry/exit movement across the doorway boundary to improve robustness under occlusion. The edge device (③) performs on-site inference and serves the live dashboard. Privacy is preserved by outputting only occupancy status and counts without identity recognition.*